SDU International Summer School

Introduction to R

Day 8 - Linear Regression

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Contents

1 Introduction

Yesterday we talked about descriptive statistics and how they are used to describe our data. We also described the relationship between two variables with a correlation coefficient, and we saw that logging a skewed variable can straighten a curved relationship.Today we will talk about a particular case of inferential statistics: modelling. In particular we will talk about: linear regression.

When our **response** variable is **continuous**, we can model our data through **linear regression**.

During this lecture we will focus on the linear regression as statistical tool that allow us to evaluate the relationship between variables, specifically between:

- One **dependent variable** (denoted by $\mathbf{y}$) called **response** and;
- One or several **independent variables** (denoted by $\mathbf{x}$) called **predictors**.

The reference for such topics is the book by McCullagh and Nelder (1989). An easier introduction and explanation is given by Agresti (2018).

1.1 Linear regression

As we mentioned before, the whole cycle of data analysis is done to answer a specific question we have in mind. For that it is very important to understand the nature of your data and define what kind of response variable and explanatory variables you have.

On Day 7 we learned that two variables can move together, and we measured how tightly with the correlation coefficient. A correlation tells us the direction and the strength of a straight-line relationship, but it does not give us the line itself. It cannot answer questions like "how many more years of life expectancy come with a given rise in income", and it treats the two variables symmetrically, as if neither depends on the other.

Regression helps with that. We choose one variable as the **response** (the thing we are trying to explain, on the y axis) and one or more variables as **predictors** (the things we explain it with, on the x axis). Then we ask for the straight line that fits the points as closely as possible. The correlation we already know and the line we are about to fit are two views of the same relationship: in a model with a single predictor, the square of the correlation is exactly the R-squared of the regression. Whereas the slope of a simple regression is the correlation rescaled by the two standard deviations,

$$\beta_1 = r\,\frac{s_y}{s_x}$$

and the intercept is whatever makes the line pass through the point of the two means, $\beta_0 = \bar{y} - \beta_1 \bar{x}$. So the correlation fixes the tilt and the two spreads fix the scale.

2 The Linear Model

2.1 The idea

A linear model with one predictor is just the equation of a straight line, plus an error term:

$$y = \beta_0 + \beta_1 x + \epsilon$$

Here y is the response, x is the predictor, β_0 is the intercept (the value of y when x is zero), β_1 is the slope (how much y changes when x goes up by one), and ϵ is the error term that soaks up everything the line does not capture. With several predictors the equation just grows more slope terms:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

R estimates the betas by **least squares**: out of all possible lines, it picks the one that makes the total squared vertical distance from the points to the line as small as possible. Those vertical distances are the **residuals**, and we will use them again when we check the model. We judge the model on the basis how likely the data would be if the model were correct.

2.2 What are we assuming:

We can always fit a linear model, but in order for it to be the correct choice we need to assume that:

- relationship between response and predictors is roughly linear (which is why we look at the scatterplots first!)
- the error term (stochastic part) ϵ follows a normal distribution with mean 0 and constant variance σ^2; i.e. the errors are scattered symmetrically around the line with no clear pattern
- the variance σ^2 of ϵ, do not depend on the covariates $x_1 + x_2 + ... + x_p$, that means their spread does not grow or shrink across the range of the predictor.

We will not expand on the theory, but we will look at two quick diagnostic plots at the end to see whether these hold.

2.3 The model in R

In R a linear model is coded as follow:

`lm(response ∼ covariate1 + covariate2, data= our.data)`

The `lm()` function fits a linear model. The `~` (tilde) reads as: "is modelled as a function of", so `life_exp ~ median_income` reads "life expectancy modelled as a function of median income".

In other words the structure of the model specified in the model as:

`response variable ~ explanatory variable`

We usually save the fitted model in an **object** so we can analyze it more later.

```
my_model <- lm(response ~ predictor1 + predictor2, data = our_data)
```

3 Simple linear regression

Before you start use any model, the best thing is to spend a substantial amount of time, right at the outset, getting to know your data and what they show!

3.1 The data

We will use real data on the health of United States counties, taken from the 2025 County Health Rankings (University of Wisconsin Population Health Institute)- Source: https://www.countyhealthrankings.org/health-data/methodology-and-sources/data-documentation.

This data includes information on 3152 counties in the US and their life expectancy, median income, percentage of smokers in the population, the share (percentage) of the population that have any post-secondary education, the percentage of population that is obese and uninsured, respectively, and the percentage of children that live in poverty . For this example we will focus on the information of the **life expectancy, median income, smokers and some_college**.

```
# setwd("C:/Users/yourname/IntroR/Day8")

# na.strings makes sure blank cells come in as NA rather than as empty text.
county_health <- read.csv("county_health.csv", na.strings = c("", "NA"))

head(county_health)
```

```
##   state          county life_exp median_income smoking some_college obesity
## 1    AL Autauga County 74.80065         68857    15.4     61.37873    38.4
## 2    AL Baldwin County 76.58011         74248    14.7     64.84788    36.8
## 3    AL Barbour County 72.70899         45298    21.9     44.07743    43.8
## 4    AL     Bibb County 72.97738         56025    21.8     43.53323    41.4
## 5    AL  Blount County 72.93617         64962    19.5     52.62711    37.3
## 6    AL Bullock County 70.96746         36204    24.9     36.31387    49.4
##   uninsured children_poverty
## 1  8.190361             17.0
## 2 10.212342             14.1
## 3 12.117042             34.8
## 4 10.835799             21.4
## 5 12.515582             16.6
## 6 11.409091             39.8
```

```
str(county_health)
```

```
## 'data.frame':    3152 obs. of  9 variables:
##  $ state           : chr  "AL" "AL" "AL" "AL" ...
##  $ county          : chr  "Autauga County" "Baldwin County" "Barbour County" "Bibb County
##  $ life_exp        : num  74.8 76.6 72.7 73 72.9 ...
##  $ median_income   : int  68857 74248 45298 56025 64962 36204 44729 51841 50390 56400 ...
##  $ smoking         : num  15.4 14.7 21.9 21.8 19.5 24.9 20.4 18 20.9 17.6 ...
##  $ some_college    : num  61.4 64.8 44.1 43.5 52.6 ...
##  $ obesity         : num  38.4 36.8 43.8 41.4 37.3 49.4 45.5 38.3 45.4 40.2 ...
##  $ uninsured       : num  8.19 10.21 12.12 10.84 12.52 ...
##  $ children_poverty: num  17 14.1 34.8 21.4 16.6 39.8 37.4 28.5 26 20.5 ...
```

The variable we want to explain is `life_exp`, the average years of life a baby born today will reach if the current mortality conditions prevail. The other columns are county characteristics we might use to explain it. Before modelling, we always look:

```
summary(county_health$life_exp)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
##   53.98   72.78   75.27   75.14   77.53   94.22     84
```

```
summary(county_health$median_income)
```

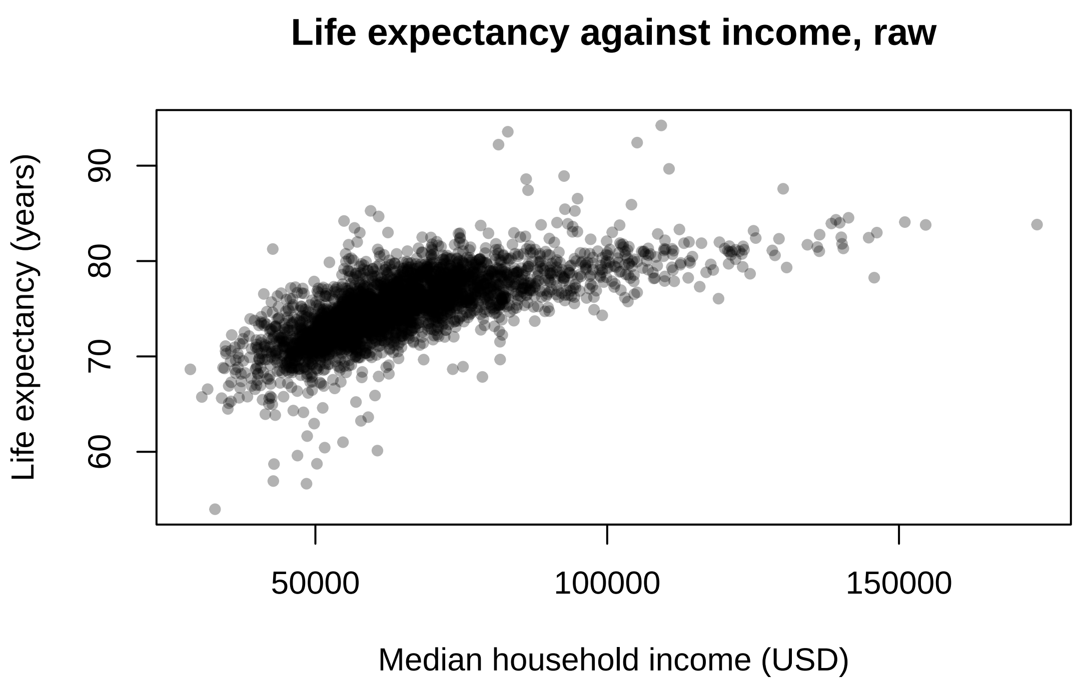
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
##   28579   54616   62851   65562   73074  173655      9
```

Notice two things. Life expectancy runs from the mid 50s to the mid 90s, a wide spread. And median income is skewed: the mean sits well above the median because a handful of very rich counties pull it up. That skew is our first clue that we may need a transformation.

3.2 Looking before we model

Income is expected to be linked to health, so we plot life expectancy against it.

```
plot(life_exp ~ median_income, data = county_health,
     xlab = "Median household income (USD)",
     ylab = "Life expectancy (years)",
     main = "Life expectancy against income, raw",
     pch = 20, col = rgb(0, 0, 0, 0.3))
```



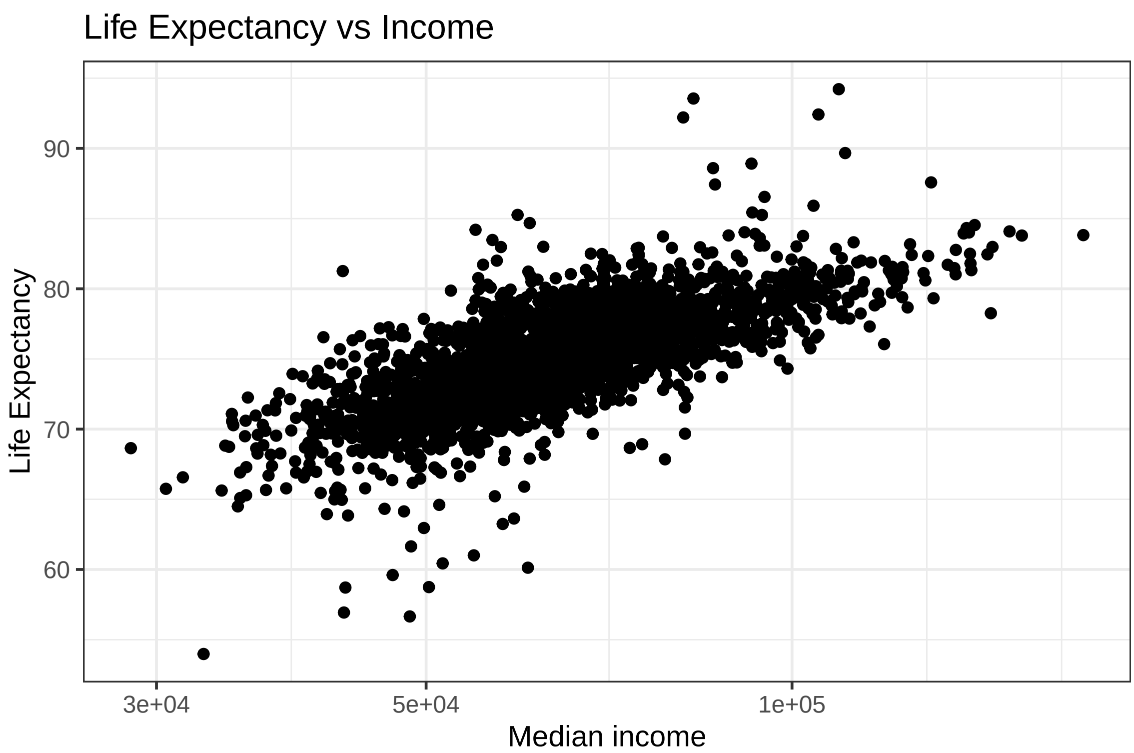
The relationship is clearly positive, but it is not a straight line. It rises steeply among poor counties and then flattens: the first few thousand dollars buy a lot of life expectancy, while the same increase among rich counties buys almost none. Does it look familiar?

3.3 Logging the predictor

Yesterday we logged a skewed variable to straighten a relationship. Here the skewed variable is the **predictor**, income, so that is the one we log.

```
library(tidyverse)

county_health |>
  ggplot(aes(median_income, life_exp))+
  geom_point()+
  labs(title="Life Expectancy vs Income",
       x= "Median income",
       y="Life Expectancy")+
  scale_x_log10()+
  theme_bw()
```



Very important:

If we have `NA` values in our dataset, the correlation will return:

```
cor(log(county_health$median_income), county_health$life_exp)
```

```
## [1] NA
```

in order to solve that, we need to add in the function the argument `use = "complete.obs"`.

```
cor(log(county_health$median_income), county_health$life_exp, use = "complete.obs")
```

```
## [1] 0.7266766
```

On the log scale the cloud lines up along a straight band, and the correlation edges up (from about 0.70 to about 0.73). A straight line is a reasonable description, so we can fit one.

A note on which variable to log. When you log the **predictor**, as we do here, the slope describes how the response changes as the predictor multiplies. If instead you had logged the **response**, the slope would describe a percentage change in the response.

3.4 Linear model and NAs

When we fit a linear model, it will just ignore the observations where there are `NAs`. However, typically we will fit more than one model and compare them, which makes specially important for us to keep the same data in all the models we do. If our `NAs` do not have a special structure and are a resonable amount (not too many), then we can remove them from the sample.

```
colSums(is.na(county_health))
```

```
##            state           county         life_exp    median_income
##                0                0               84                9
##          smoking     some_college          obesity        uninsured
##                8                9                8                9
## children_poverty
##                9
```

```
# We have 84 counties with no info in our interest variable (y)
# for our model, we can remove them since they provide no information
# same with smoking and some college
model_data <- drop_na(county_health)
# remember we can also just remove the columns

# confirm the dimension
dim(model_data)
```

```
## [1] 3060    9
```

3.5 Fitting the model

When a preliminary explanatory data analysis is done, we think about model choice:

1. Which model is more appropriate? What is the nature of my response variable and which explanatory variables do I want to include?

2. Try to use the simplest model that is appropriate for your data!

Given our common knwoledge and the scatterplot above, it seems reasonable to assume a positive linear relationship between life expectancy and the (log) median income of a county. The poorer the county is, the lower the life expectancy is.

```
m_simple <- lm(life_exp ~ log(median_income), data = model_data)
summary(m_simple)
```

```
##
## Call:
## lm(formula = life_exp ~ log(median_income), data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.4223  -1.4520  -0.0908   1.4042  15.4559
```

```
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        -48.7818     2.1189  -23.02   <2e-16 ***
## log(median_income)  11.2022     0.1915   58.49   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.49 on 3058 degrees of freedom
## Multiple R-squared:  0.5281, Adjusted R-squared:  0.5279
## F-statistic:  3422 on 1 and 3058 DF,  p-value: < 2.2e-16
```

NB! After you have run your model, document carefully what you have done, and explain all the steps you took. That way, you should be able to understand what you did and why you did it, when you return to the analysis after some time.

3.6 Interpretation of the summary

The `summary()` of a fitted model looks dense the first time, so here is every block, in order.

- **Call** simply repeats the model you fitted, so you can tell models apart later.
- **Residuals** are the leftover vertical distances from each point to the line. You want them roughly symmetric around zero (the median near zero, the min and max not wildly lopsided), which suggests the line sits fairly through the middle of the cloud of points.
- **Coefficients, Estimate** are the betas. The `(Intercept)` is the predicted life expectancy when `log(median_income)` is zero (not meaningful on its own here, since no county has an income of one dollar). The second row is the slope on `log(median_income)`, roughly 11.2.
- **Std. Error** is how much each estimate would wobble if we drew another sample of counties. Smaller is better.
- **t value** is the estimate divided by its standard error, and **Pr(>|t|)** is the p-value: the probability of seeing a coefficient this large if the true effect were zero. The stars flag small p-values. Here both p-values are far below 0.001, so the relationship is very unlikely to be chance.
- **Residual standard error** is the typical size of a residual, in years of life expectancy: roughly how far off a prediction tends to be.
- **Multiple R-squared** is the share of the variation in life expectancy that the model explains. Here it is about 0.53, so income alone accounts for a bit over half of why counties differ in life expectancy. **Adjusted R-squared** is the same idea with a small penalty for model size, which matters once we compare models below.
- **F-statistic** and its p-value test whether the model as a whole explains anything. Far from one, with a tiny p-value, means yes.

3.7 Where the slope comes from

The slope we just estimated is not new machinery. It is exactly the formula:

$$cor_{x,y} * \frac{s_y}{s_x}$$

would have handed us from the correlation and the two standard deviations, as long as we use the same rows the model used.

```r
r  <- cor(log(model_data$median_income), model_data$life_exp)
sy <- sd(model_data$life_exp)
sx <- sd(log(model_data$median_income))

r * sy / sx          # yesterday's formula: r * sd(y) / sd(x)
```

```
## [1] 11.20223
```

```r
m_simple$coefficients  # model's coefficients [1] is intercept and [2] is beta
```

```
##        (Intercept) log(median_income)
##          -48.78177           11.20223
```

The two match. The regression has taken the correlation you already knew, scaled it by the ratio of the two spreads, and drawn the line through the point of the means. What `lm()` adds is the standard errors, the significance, and the ease of extending to several predictors at once.

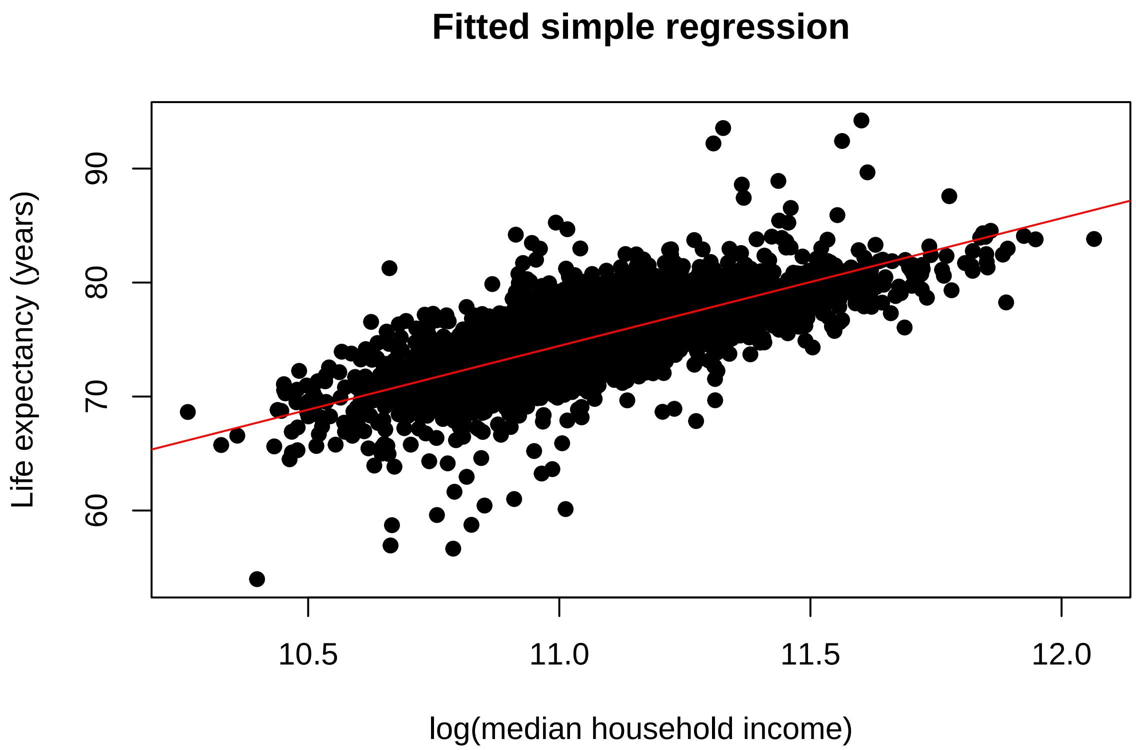
3.7.1 A note on interpreting a log slope

The slope on `log(median_income)` is about 11.2. Because the predictor is logged, read it as a change per multiplication of income, not per dollar. A friendly interpretation uses doubling: doubling income raises `log(median_income)` by `log(2) = 0.69`, about 0.69, so the model links a doubling of median income with roughly `0.69 * 11.2`. So about 8 more years of life expectancy are gained every time income is doubled. Same eight years whether it takes a county from 20,000 to 40,000 dollars or from 80,000 to 160,000, so in absolute dollars the gains from increasing income shrink as counties get richer. If instead we wanted to think about income changes by `10%`, we would look at the `log(1.10)=0.095` and say that for every `10%` of increase in median income would increase life expectancy by `0.095 * 11.2 = 1.07` years.

3.8 Plot the estimated regression line

A graphical view of the fitted model is obtained by plotting the regression line on the actual data.

```r
plot(log(county_health$median_income), county_health$life_exp,
     xlab = "log(median household income)",
     ylab = "Life expectancy (years)",
     main = "Fitted simple regression",
     pch = 19)
abline(m_simple, col="red")
```



The line runs through the middle of the band. Points scatter above and below it, which is the residual variation the model does not explain, and which the next step tries to reduce.

4 Multiple regression

4.1 Why add more predictors

Income is not the only thing that makes a county live longer or shorter. Counties also differ in how much people smoke and in how educated they are, and both potentially matter for health. Two reasons push us to put them in the same model. First, we may simply explain more of the variation. Second, and more subtly, part of what looked like an income effect might really be a smoking or education effect travelling alongside income, because poorer counties also tend to smoke more and have less schooling. A multiple regression lets us ask: what is the association of income with life expectancy **among counties that smoke the same amount and have the same education**?

4.2 Fitting and reading the model

We add predictors with `+`.

```
m_multi <- lm(life_exp ~ log(median_income) + smoking + some_college,
              data = model_data)
summary(m_multi)
```

```
##
## Call:
## lm(formula = life_exp ~ log(median_income) + smoking + some_college,
```

```
##     data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6410  -1.4174  -0.0884   1.3435  14.4162
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         6.896708   3.054851   2.258    0.024 *
## log(median_income)  6.465776   0.266330  24.277  < 2e-16 ***
## smoking            -0.301554   0.016706 -18.051  < 2e-16 ***
## some_college        0.036364   0.005035   7.222 6.45e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.288 on 3056 degrees of freedom
## Multiple R-squared:  0.6019, Adjusted R-squared:  0.6015
## F-statistic:  1540 on 3 and 3056 DF,  p-value: < 2.2e-16
```

4.3 Interpreting each coefficient

Every slope in a multiple regression is read the same way, with one crucial phrase added: **holding the other predictors constant**.

- **log(median_income)**, now about 6.5. Among counties with the same smoking rate and the same share of college-educated adults, higher income is still linked to higher life expectancy, but the association is smaller than before.
- **smoking**, about -0.30. Because we put smoking on a percentage scale, this says that each additional percentage point of adults who smoke is linked with about a third of a year less life expectancy, holding income and education constant. It is strongly significant, and it is the biggest single mover here.
- **some_college**, about 0.036. Holding income and smoking constant, more college educated population is linked with slightly higher life expectancy. The effect is small once smoking is in the model, which itself is worth noticing: much of what education does for health seems to run through behaviours like smoking.

The adjusted R-squared has risen to about 0.60, so the three predictors together explain more than income did alone.

4.4 When a coefficient changes: confounding

The number to circle is income's slope. On its own it was about 11.2. With smoking and education in the model it falls to about 6.5, nearly half. Nothing about income changed. What changed is that we stopped crediting income for differences that were really about smoking and schooling. Poorer counties smoke more, and smoking shortens lives, so a simple income-only model quietly folded some of smoking's harm into the income coefficient. Adding smoking pulls that borrowed effect back out.

This is the practical face of a warning: correlation is not causation, and a third variable can inflate, shrink, or even reverse an association. It is also why "controlling for" other variables is the everyday work of applied regression. It does not prove that income causes health, but it gives a more honest estimate of income's own association by removing the most obvious competing explanations.

5 Comparing models

We now have two models of the same response. Which should we report? We need a fair way to compare them.

5.1 R-squared always rises

Plain R-squared can never fall when you add a predictor, even a useless one, because the model can always use the extra freedom to fit the noise a little better. So a bigger R-squared for a bigger model proves nothing on its own. Two tools correct for this.

Adjusted R-squared takes R-squared and subtracts a penalty for each predictor. It only rises if a new predictor earns its place. Our simple model had adjusted R-squared about 0.53; the multiple model has about 0.60, a real improvement.

AIC (the Akaike Information Criterion) scores a model on how well it fits while penalising it for the number of terms it uses. Lower is better. Unlike R-squared, AIC has no natural scale, so the number only means something in comparison to another model's AIC on the same data.

```r
AIC(m_simple, m_multi)
```

```
##          df      AIC
## m_simple  3 14272.30
## m_multi   5 13755.89
```

The multiple model's AIC is lower by roughly 500, a large gap.

5.2 An ANOVA test of the improvement

Adjusted R-squared and AIC each give a single score per model. A third tool asks the question directly: do the extra predictors, taken together, improve the fit by more than chance would give? For two **nested** models (the simple model is the multiple model with two predictors removed), `anova()` runs exactly that test.

```r
anova(m_simple, m_multi)
```

```
## Analysis of Variance Table
##
## Model 1: life_exp ~ log(median_income)
## Model 2: life_exp ~ log(median_income) + smoking + some_college
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1   3058 18968
## 2   3056 16001  2    2966.4 283.27 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This is an analysis of variance: it splits the leftover variation and tests whether the drop in residual variation from adding smoking and education is larger than we would expect by chance. The F value is large and its p-value is tiny, so the answer is yes, the two predictors earn their place. All three tools now agree, and the multiple model is the one we would report.

Because both models were fitted on the same `model_data`, `anova()` runs cleanly. It is fussier than `AIC()` about this: if the two models had different numbers of rows, `anova()` would stop with an error rather than hand back a wrong answer. That strictness is a useful safety net.

5.3 Model checking

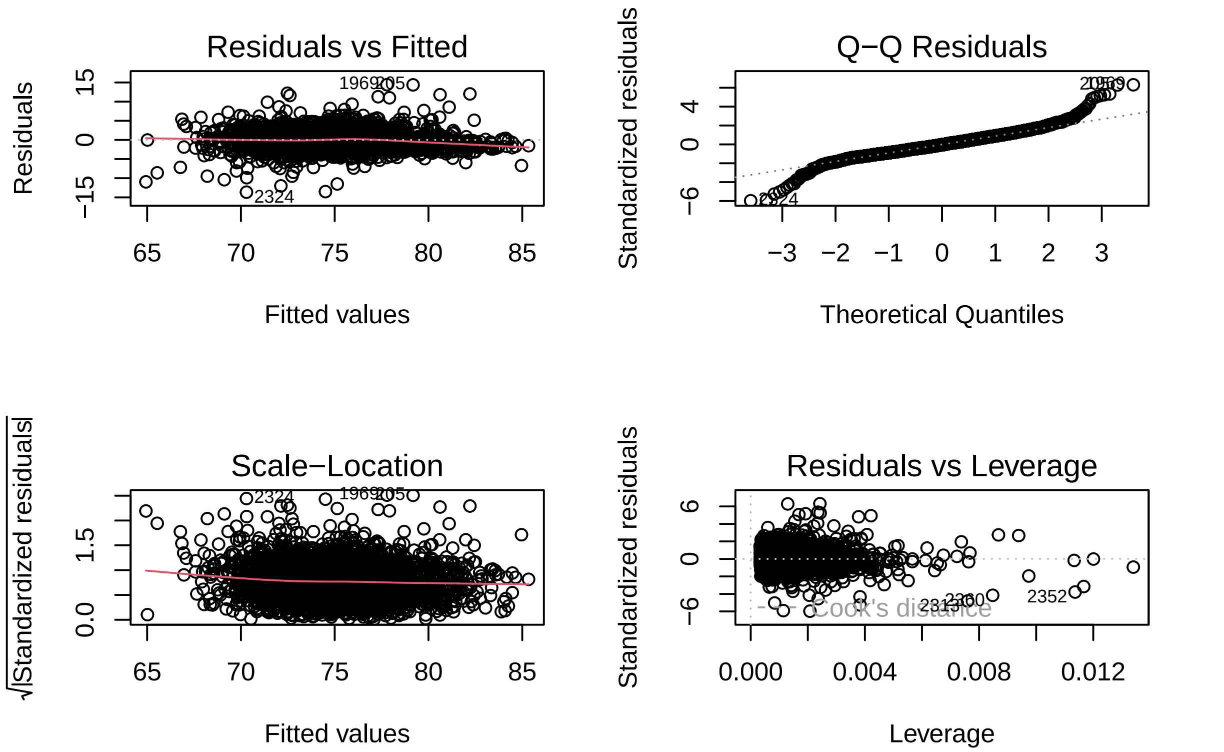
After fitting a model to data we need to investigate how well the model describes the data. The possible influence of outliers and checking if assumptions made were fulfilled (e.g. constant variance and normal errors). R has different tools for doing this.

The simplest and most well known are the estimated residuals, i.e., the differences between the observed values of the response and the fitted values of the response. In essence these residuals estimate the error terms in the linear regression.

Useful plots of these residuals are obtained with `plot(model)`.

```
par(mfrow=c(2,2)) # to split the screen 2 by 2

plot(m_multi)
```




```
par(mfrow=c(1,1)) # go back to normal
```

The **first graph** (top left) shows residuals (y axis) against fitted values (x axis). This plot should look like a shapeless band around zero, with no funnel and no curve. A curve would say the relationship is not straight; a funnel would say the spread of errors changes across the range.

The **second plot** (top right) shows the normal qqnorm plot which should be a straight line if the errors are normall y distributed. Again, the present example looks fine. If the pattern were

S-shaped or banana-shaped, we would need to fit a different model to the data.

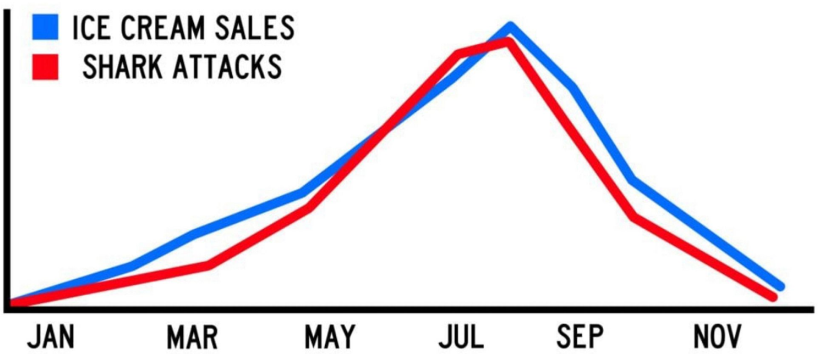
The **third plot** (bottom left) is a repeat of the first, but on a different scale; it shows the square root of the standardized residuals (where all the values are positive) against the fitted values. If there was a problem, such as the variance increasing with the mean, then the points would be distributed inside a triangular shape, with the scatter of the residuals increasing as the fitted values increase. The red line would then show a pronounced upward trend. Here we do not see a pronounced pattern.

The **fourth** and final plot (bottom right) shows standardized residuals as a function of leverage, along with Cook's distance for each of the observed values of the response variable. The Cook´s distance indicates the difference between the fitted values and the values one would have obtained, if the respective case was excluded. All distances should be of about equal magnitude; if not, then there is reason to believe that the respective case(s) biased the estimation of the regression coefficients (outliers?). The point of this plot is to highlight those y values that have the biggest effect on the parameter estimates and to help us identifying possible outliers.

6 CAUTION: Correlation does not imply causation

It is a common statistical mistake to assume that when two variables are correlated, that means that one must cause the other. However, causation needs to be tested separately.

Let's think for example about the timing of shark attacks. There is evidence that shark attacks happen more often from May to September. We also observe that from May to September the ice cream sales are higher than the rest of the year. Can we conclude then, that ice creams cause the sharks to attack? NO!



Although ice cream sales and shark attacks might have a strong correlation, one is not causing the other. So any time that you see one variable correlated to another remember: this does not mean causation.

6.1 Video

The danger of mixing up causality and correlation!

https://www.youtube.com/watch?v=8B271L3NtAw

6.2 How to measure causality?

Causality is a cause-and-effect relationship between one independent variable (x) and one dependent variable (y).

The classical requirements for detecting a causal link:

1. Existence of correlation between x and y.
2. Theoretically plausible causal mechanism: Is there a credible causal mechanism that connects x and y?
3. Time order: x comes before y in time. Can we rule out the possibility that y could cause x?
4. Elimination of alternative explanations (control for third variable).

7 Summing up

- The linear model is a statistical tool that allows us to investigate the relationship between continuous variables.
- Regression puts a line through the cloud that a correlation only summarised, and it distinguishes a response from its predictors.
- A linear model has assumptions that should be met for it to be an acceptable model.
- A good fit for a linear model implies correlation between response and predictors.
- The fitted slope is the correlation times sd(y) over sd(x).
- `lm(response ~ predictor, data = ...)` fits the model; `summary()` reads it.
- Log a skewed predictor before fitting; then read its slope as a change per multiplication of the predictor.
- In a multiple regression, each slope is read holding the others constant, and a coefficient can change a lot when you add the right third variable. That is confounding, and it is why we control for things.
- Compare models with adjusted R-squared, AIC, and an `anova()` F-test, never just plain R-squared, and only ever on the **same observations**.
- Correlation is not causation. A well-controlled model is more honest, not a proof.

8 Glossary

The functions from today, grouped by what you use them for.

Fitting and reading a model

#	Function	What it does
1	`lm()`	Fits a linear model from a `response ~ predictor` formula
2	`summary()`	Prints the coefficients, R-squared, and significance of a fitted model
3	`coef()`	Returns just the estimated intercept and slopes
4	`confint()`	Returns confidence intervals for the coefficients

Using a fitted model

#	Function	What it does
5	`predict()`	Uses the model to predict the response for new or existing rows
6	`residuals()`	Returns the leftover distances from points to the line
7	`abline()`	Draws a fitted one-predictor line onto an existing scatter plot
8	`plot()`	On a model object, draws the diagnostic residual plots

Transforming and comparing

#	Function	What it does
9	`AIC()`	Scores models for comparison; lower is better, same data only
10	`anova()`	Tests whether extra predictors jointly improve a nested model

9 References

Agresti, A. (2018). Statistical Methods for the Social Sciences (fifth edition). Pearson.

Dalgaard, P. (2008). Introductory Statistics with R (second edition). Springer.

University of Wisconsin Population Health Institute (2025). County Health Rankings and Roadmaps. Available at https://www.countyhealthrankings.org.

10 Exercises

Open the dataset `county_health.csv` again, but this time work with columns the lecture did not use: `children_poverty` (the percentage of children living in poverty), `obesity` (the percentage of adults with obesity), and `uninsured` (the percentage of adults without health insurance). The response is still `life_exp`.

- Produce a scatter plot of `life_exp` against `children_poverty`, and calculate their correlation coefficient. Describe the direction and strength of the relationship.
- Try a log transformation of `children_poverty`, plot it again, and recompute the correlation. Did it change much? What does that tell you about whether this variable needs transforming?
- Fit a simple linear model predicting `life_exp` from `children_poverty`. Add the regression line to your scatter plot. Interpret the slope and its significance. On average, how much does life expectancy differ for each extra percentage point of children in poverty?
- Fit a second model that adds `obesity` and `uninsured` as predictors. Interpret each coefficient, holding the others constant. Did the coefficient on `children_poverty` change compared with the simple model? What does the change suggest?
- Compare the two models using adjusted R-squared and AIC. Which would you report, and why? Use `anova()` to test whether `obesity` and `uninsured` together significantly improve the model over the poverty-only model. What do the F value and its p-value tell you?

Remember: both comparisons only make sense if the two models were fitted on the same counties (observations!). Why does this matter?